Team SimGym

Track 4 : Ai Customer Support Agent for Commerce

This document explains what we are building, for whom we are building it, and why it matters. It focuses on problem framing, product decisions, scope limitations, and tradeoff handling rather than marketing language.

1. The Problem We Are Solving and Why It Matters

Traditional online shopping platforms often create problems for users because product discovery, customer support, order tracking, return handling, and purchase decisions are separated into different sections. Users must manually search for products, compare options, track orders, and understand return policies without instant support.

This creates a poor user experience, especially for users who want quick answers before making a purchase. Many users leave the platform when they cannot find information easily.

This project solves that problem by combining an e-commerce storefront with an AI-Based Shopping Assistant Chatbot. The chatbot provides instant help for product recommendations, cart summary, order tracking, return support, and shopping-related questions.

This matters because faster support improves customer satisfaction, reduces confusion, and increases the chances of successful purchases.

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2. Target Users and Current Experience

Target Users

The primary target users are:

- Online buyers/customers
- Small and medium e-commerce businesses

Current User Experience

In many traditional systems:

- Users search manually through large product lists

- Product recommendations are limited
- Customer support is slow or unavailable
- Return and order tracking information is difficult to find
- Users must navigate multiple pages for simple tasks

This causes frustration and reduces shopping efficiency.

Our system improves this by offering both shopping and support in one place.

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3. What We Decided to Build

We decided to build an AI-Based E-Commerce Storefront application using React with an integrated intelligent chatbot assistant.

Core Features Built

- User login and profile storage using Local Storage
- Product listing through Storefront component
- Add to Cart functionality
- Wishlist management
- Cart quantity management
- AI-Based Shopping Assistant Chatbot
- Product recommendation support
- Order tracking widget
- Return initiation support
- Cart summary display
- Quick suggestion buttons for common customer queries

Core User Journey

1. User opens the application
2. User logs in and profile is stored
3. User browses available products
4. User interacts with chatbot for support
5. Chatbot helps with recommendations, tracking, returns, and cart summary

6. User completes shopping with better decision-making and faster support

This journey reduces friction and improves the complete shopping process.

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4. Key Product Decisions and Reasoning

Decision 1: Integrated Chatbot Instead of Separate Support Page

Reason: Users need immediate help without leaving the shopping page. A floating chatbot improves accessibility and reduces navigation time.

Decision 2: Local Storage for User Login

Reason: Local storage keeps user information available even after page refresh without requiring backend authentication for the prototype stage.

Decision 3: Suggestion-Based Chat Prompts

Reason: Many users do not know what to ask. Quick suggestion buttons improve chatbot usability and encourage interaction.

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5. What We Chose NOT to Build (Scope Decisions)

To keep the project practical and focused, some features were intentionally excluded.

I chose this track because, while browsing different e-commerce platforms and reading customer reviews, I noticed that many users face repeated problems during online shopping. Most complaints were related to difficulty contacting the service, lack of proper recommendations, slow customer support, confusion about return policies, and poor communication during the buying process. Many users leave the platform without completing their purchase because they cannot get quick answers to simple questions. This showed that the problem is not only about selling products, but also about improving the complete customer experience. Based on these observations, I decided to build an AI-Based E-Commerce Storefront with an intelligent shopping assistant chatbot that can provide instant support, simplify decision-making, and make online shopping faster, smoother, and more user-friendly.

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6. Tradeoffs Faced and How They Were Resolved

Tradeoff 2: Full Backend vs Frontend Prototype

Challenge: A complete backend system requires more time and database management.

Resolution: We prioritized frontend functionality and user experience using React and local storage for faster delivery.

Tradeoff 3: More Features vs Better Stability

Challenge: Adding too many features could reduce project quality and increase bugs.

Resolution: We focused on fewer high-value features like cart, and chatbot support to ensure better performance and reliability.